

Homework 6

1) Read Section 4.1, 4.2 & 4.3 & 4.5 of Nakahara to learn about homotopy groups.

Then, compute the homotopy groups of the following spaces:

a) compute the homotopy groups of S^2 .

b) Compute the homotopy groups of a torus: $S^1 \times S^1$

2) Show that homotopy group $\pi_r(M)$ is Abelian for $r > 1$.